

1. A power company wants to predict the electricity demand in different regions based on past usage, temperature, and time of year. What type of machine learning approach would be suitable? Outline the steps to build the model.

Problem Domain:

Step 1: Machine learning domain as past usage, temperature are in numbers

Step 2: Supervised – as requirement is clear (labeled)

Step 3: classification (as we can categorize electricity demand in high, medium, low)

Work flow:

1. Load the electricity_usage dataset
2. Check for missing values and handle it
3. Check for categorical values and handle it
4. Do the data preprocessing if required
5. Split the input & output
6. Split train and test data
7. Model creation (logistic regression, KNN, Naïve bayes etc..)
8. Model testing
9. Evaluate the model using confusion matrix
10. Finalize the best model based on comparison of results, use case etc..

2. An e-commerce platform wants to identify and filter out fake product reviews based on user behavior, review patterns, and sentiment analysis. What type of machine learning approach would be suitable? Outline the steps to build the model.

Problem Domain:

Step 1: Machine learning domain as no of reviews, user logins are in numbers

Step 2: Supervised – as requirement is clear (labeled)

Step 3: classification (as we can categorize Review – fake or genuine)

Work flow:

1. Load the e-commerce_reviews dataset
2. Check for missing values and handle it
3. Check for categorical values and handle it

4. Do the data preprocessing if required
 5. Split the input & output
 6. Split train and test data
 7. Model creation (logistic regression, KNN, Naïve bayes etc..)
 8. Model testing
 9. Evaluate the model using confusion matrix
 10. Finalize the best model based on comparison of results, use case etc..
3. A city traffic department wants to estimate traffic congestion levels based on GPS data, road construction reports, and weather conditions. What type of machine learning approach would be suitable? Outline the steps to build the model.

Problem Domain:

Step 1: Machine learning domain as gps data, weather conditions are in numbers

Step 2: Supervised – as requirement is clear (labeled)

Step 3: classification (as we can categorize traffic congestion levels – high, medium, low)

Work flow:

1. Load the city_traffic dataset
 2. Check for missing values and handle it
 3. Check for categorical values and handle it
 4. Do the data preprocessing if required
 5. Split the input & output
 6. Split train and test data
 7. Model creation (logistic regression, KNN, Naïve bayes etc..)
 8. Model testing
 9. Evaluate the model using confusion matrix
 10. Finalize the best model based on comparison of results, use case etc..
4. A bank wants to determine whether a loan applicant should be approved or rejected based on income, credit history, and previous loan repayment behavior. What type of machine learning approach would be suitable? Outline the steps to build the model.

Problem Domain:

Step 1: Machine learning domain as income, credit history are in numbers

Step 2: Supervised – as requirement is clear (labeled)

Step 3: classification (as we can categorize loan applicant – approved / rejected)

Work flow:

1. Load the bank_loan dataset
2. Check for missing values and handle it
3. Check for categorical values and handle it
4. Do the data preprocessing if required
5. Split the input & output
6. Split train and test data
7. Model creation (logistic regression, KNN, Naïve bayes etc..)
8. Model testing
9. Evaluate the model using confusion matrix
10. Finalize the best model based on comparison of results, use case etc..

5. A factory wants to automatically detect defective products using images from a quality control camera. What type of machine learning approach would be suitable? Outline the steps to build the model.

Problem Domain:

Step 1: Machine learning domain

Step 2: Unsupervised – as requirement is unclear (unlabeled)

Step 3: clustering (as we can cluster product images – outliers of the cluster are the defective products)

Work flow:

1. Load the factory_product_dataset
2. Select the important features of the dataset which major impact the detection of defective product – X input
3. Data preprocessing
4. Finding no of clusters (dendrogram, elbow method etc..)
5. Model creation & prediction using (k means, agglomerative, dbscan etc..)
6. Merge the new cluster group and save it in csv file
7. Plot the cluster using seaborn – to visualize it
8. Finalize the models by comparing

6. An agricultural company wants to predict the best crops to grow based on soil composition, rainfall, and past harvest data. What type of machine learning approach would be suitable? Outline the steps to build the model.

Problem Domain:

Step 1: Machine learning domain as gps data, weather conditions are in numbers

Step 2: Supervised – as requirement is clear (labeled)

Step 3: classification (as we can categorize crops – wheat, rice, greens, etc..)

Work flow:

1. Load the agricultural dataset
2. Check for missing values and handle it
3. Check for categorical values and handle it
4. Do the data preprocessing if required
5. Split the input & output
6. Split train and test data
7. Model creation (logistic regression, KNN, Naïve bayes etc..)
8. Model testing
9. Evaluate the model using confusion matrix
10. Finalize the best model based on comparison of results, use case etc..

7. A business wants to analyze customer responses to different types of advertisements and promotions to determine the most effective marketing strategy. What type of machine learning approach would be suitable? Outline the steps to build the model.

Problem Domain:

Step 1: Machine learning domain

Step 2: Unsupervised – as requirement is unclear (unlabeled)

Step 3: clustering (as we can cluster – customer ads based on response)

Work flow:

1. Load the customer dataset
2. Select the important features of the dataset which major impact the detection of defective product – X input
3. Data preprocessing

4. Finding no of clusters (dendrogram, elbow method etc..)
5. Model creation & prediction using (k means, agglomerative, dbscan etc.)
6. Merge the new cluster group and save it in csv file
7. Plot the cluster using seaborn – to visualize it
8. Finalize the models by comparing

8. A software company wants to build a system that can analyze source code and predict whether a particular piece of code is likely to contain a bug. What type of machine learning approach would be suitable? Outline the steps to build the model.

Problem Domain:

Step 1: Machine learning domain as system code are in numbers

Step 2: Supervised – as requirement is clear (labeled)

Step 3: classification (as we can categorize bugs – attribute error, syntax error, value error, etc..)

Work flow:

1. Load the software_sourceCode dataset
2. Check for missing values and handle it
3. Check for categorical values and handle it
4. Do the data preprocessing if required
5. Split the input & output
6. Split train and test data
7. Model creation (logistic regression, KNN, Naïve bayes etc..)
8. Model testing
9. Evaluate the model using confusion matrix
10. Finalize the best model based on comparison of results, use case etc..

9. A fitness app wants to recommend personalized workout plans for users based on their exercise history, fitness level, and preferences. What type of machine learning approach would be suitable? Outline the steps to build the model.

Problem Domain:

Step 1: Machine learning domain

Step 2: Unsupervised – as requirement is unclear (unlabeled)

Step 3: clustering (as we can cluster – personalized workouts based on their exercise history, fitness level etc..)

Work flow:

1. Load the user_fitness dataset
2. Select the important features of the dataset which major impact the detection of defective product – X input
3. Data preprocessing
4. Finding no of clusters (dendrogram, elbow method etc..)
5. Model creation & prediction using (k means, agglomerative, dbscan etc.)
6. Merge the new cluster group and save it in csv file
7. Plot the cluster using seaborn – to visualize it
8. Finalize the models by comparing

10. A social media platform wants to detect fake accounts by analyzing user activity, posting patterns, and interactions with other users. Give steps to Achieve.

Problem Domain:

Step 1: Machine learning domain as user interaction nos, posting patterns nos are in numbers

Step 2: Supervised – as requirement is clear (labeled)

Step 3: classification (as we can categorize user accounts – fake or genuine)

Work flow:

1. Load the social_media_users dataset
2. Check for missing values and handle it
3. Check for categorical values and handle it
4. Do the data preprocessing if required
5. Split the input & output
6. Split train and test data
7. Model creation (logistic regression, KNN, Naïve bayes etc..)
8. Model testing
9. Evaluate the model using confusion matrix
10. Finalize the best model based on comparison of results, use case etc..